Assignment 12

Total = 10 Marks

Multiple Choice Questions

Question 1: An analyst wants to predict the target outcome of a particular event using three continuous features. Let x_1, x_2, and x_3 be the observed magnitudes of these three features. The following 12 historical observations are available, categorized into Class A and Class B:

Observation	x_1	x_2	x_3	Class
1	14	10	6	A
2	8	8	5	B
3	11	9	6	A
4	6	8	5	B
5	13	8	5	A
6	9	7	5	B
7	12	11	6	A
8	10	6	6	B
9	11	8	5	A
10	7	5	2	B
11	11	10	4	A
12	9	5	4	B

For a new event, the feature magnitudes are $X = (x_1, x_2, x_3) = (10, 8, 5)$. Using the K-Nearest Neighbours (KNN) classification algorithm with $K = 7$ and Euclidean distance, what is the predicted class of the new event?

(a) Class B, because of 2 Class A and 5 Class B nearest neighbors
(b) Class B, because of 3 Class A and 4 Class B nearest neighbors
(c) Class A, because of 4 Class A and 3 Class B nearest neighbors
(d) Class A, because of 5 Class A and 2 Class B nearest neighbors

[3 *Mark*]

Correct answer: (c) Class A, because of 4 Class A and 3 Class B nearest neighbors

Detailed Solution:

The Euclidean distance from the target $X = (x_1, x_2, x_3) = (10,8,5)$ to each observation is calculated in the following table. The ranks are also assigned based on the values of Euclidean Distance.

$$\text{Euclidean distance } (d) = \sqrt{(X_1 - x_1)^2 + (X_2 - x_2)^2 + (X_3 - x_3)^2}$$

Observation	x_1	x_2	x_3	Class	Euclidean distance (d)	Rank
1	14	10	6	A	21	11
2	8	8	5	B	4	4
3	11	9	6	A	3	3
4	6	8	5	B	16	10
5	13	8	5	A	9	7
6	9	7	5	B	2	2
7	12	11	6	A	14	9

8	10	6	6	B	5	5
9	11	8	5	A	1	1
10	7	5	2	B	27	12
11	11	10	4	A	6	6
12	9	5	4	B	11	8

From the table, 7 (as $K = 7$) nearest neighbour observations are 9, 6, 3, 2, 8, 11 and 5 and their corresponding classes are A, B, A, B, B, A and A. So, the no. of A votes are 4 and no. of B votes are 3. The majority no. of votes is for the class A.

Hence, the correct answer is: Class A, because of 4 Class A and 3 Class B nearest neighbors.

Question 2: The main role of PCA in the study with multi-dimensional predictor data is to:

(a) Increase the dimensionality of the predictor dataset
(b) Reduce the dimensionality of the predictor dataset
(c) Retain all original predictors without transformation
(d) None of these

[1 *Mark*]

Correct answer: (b) Reduce the dimensionality of the predictor dataset

Detailed Solution:

Principal Component Analysis (PCA) transforms a large set of potentially correlated predictors into a smaller number of uncorrelated principal components while retaining most of the important variability in the original data. This reduces the dimensionality of the predictor dataset and redundancy, if any, and makes subsequent modelling more efficient.

Question 3: When developing a data-driven forecasting model, a K-fold cross-validation technique is often employed. If a researcher chooses $K = 5$, what percentage of the total dataset is used for *training* the model in each individual fold?

(a) 5%
(b) 10%
(c) 20%
(d) 80%

[1 *Mark*]

Correct answer: (d) 80%

Detailed Solution:

In a 5-fold cross-validation, the dataset is divided into 5 equal parts. In each iteration, 4 parts (80%) are used for training, and the remaining 1 part (20%) is used for testing.

Question 4: Which specific component within an LSTM cell is responsible for deciding what existing information from the previous cell state should be discarded?

(a) The Input Gate
(b) The Output Gate
(c) The Forget Gate
(d) None of these

[1 *Mark*]

Correct answer: (c) The Forget Gate

Detailed Solution:

The forget gate looks at the previous hidden state and the current input, and outputs a number between 0 and 1 for each number in the cell state. A value of 0 means "completely get rid of this," while a 1 means "completely keep this," thereby controlling what information is discarded.

Question 5: The Nash-Sutcliffe Efficiency (NSE) is a widely used metric to assess the model performance. Suppose you are evaluating a simplified forecasting model. The observed values, y_{obs} are: $4, 6, 8$ and the predicted values, y_{pred} are: $5, 6, 7$ respectively. The Nash-Sutcliffe Efficiency (NSE) for this model is:

(a) 0.65
(b) 0.75
(c) 0.85
(d) 0.95

[2 *Mark*]

Correct answer: (b) 0.75

Detailed Solution:

NSE is defined as:

$$NSE = 1 - \frac{\sum(y_{obs} - y_{pred})^2}{\sum(y_{obs} - \overline{y_{obs}})^2}$$

Step – 1: Calculating the mean of the observed values:

$$\overline{y_{obs}} = \frac{4 + 6 + 8}{3} = \frac{18}{3} = 6$$

Step – 2: Calculating the numerator (Sum of Squared Errors):

$$\sum(y_{obs} - y_{pred})^2 = (4 - 5)^2 + (6 - 6)^2 + (8 - 7)^2$$

$$= (-1)^2 + (0)^2 + (1)^2 = 1 + 0 + 1 = 2$$

Step – 3: Evaluating the denominator (Total Sum of Squares):

$$\sum(y_{obs} - \overline{y_{obs}})^2 = (4 - 6)^2 + (6 - 6)^2 + (8 - 6)^2$$

$$= (-2)^2 + (0)^2 + (2)^2 = 4 + 0 + 4 = 8$$

Hence, the NSE is:

$$NSE = 1 - \frac{2}{8} = 1 - 0.25 = 0.75$$

Question 6: A multi-class Deep Learning model is developed to classify a target event into three categories: Class A, Class B, and Class C. The model is evaluated using a testing dataset of 100 observations. The resulting confusion matrix is given below:

	Predicted Class A	**Predicted Class B**	**Predicted Class C**
Observed Class A	10	5	0
Observed Class B	10	40	5
Observed Class C	10	5	15

The overall accuracy for this model is:

(a) 0.65
(b) 0.75
(c) 0.85
(d) 0.95

[2 *Mark*]

Correct answer: (a) 0.65

Detailed Solution:

The overall accuracy of this classification model is defined as the ratio of correctly predicted observations to the total observations. From the given data:

True Class A (Observed Class A and Predicted Class A) = 10

True Class B (Observed Class B and Predicted Class B) = 40

True Class C (Observed Class C and Predicted Class C) = 15

Now, calculating the sum of correct predictions:

$$\text{Correct} = 10 + 40 + 15 = 65$$

Then, evaluating the total number of observations:

$$\text{Total} = 10 + 5 + 0 + 10 + 40 + 5 + 10 + 5 + 15 = 100$$

Hence, the overall accuracy is:

$$\text{Accuracy} = \frac{\text{Correct}}{\text{Total}} = \frac{65}{100} = 0.65$$